LESSON PLAN

MOCKUP SQUADRON ENVIRONMENT (MSE) (24)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	07
Total No. of Lesson Plans	21	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Demonstration of construction and explanation on operation of HEIU.
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Demonstration of construction and explanation on operation of HEIU. Safety precautions to be observed while working on HEIU.
5. Specific Objectives	24.6.1.1 Trainee will be able to explain the operation of HEIU.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Demonstration of HEIU construction and operation; Safety precautions; Instructor supervision
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the full form of HEIU?
(b) Where is the HEIU typically located in the aircraft?
(c) What is the primary function of the HEIU in the hydraulic system?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Introduction to HEIU components • Construction details of HEIU • Internal mechanism overview • Fluid flow path explanation • Component identification	L & Ds	PPT, IB	15 min
(b)	• Demonstration of HEIU operation • Safety precautions during work • Operational sequence explanation • Pressure regulation mechanism • Troubleshooting common issues	L & Ds	PPT, PDF	20 min
(c)	• Review of safety protocols • Summary of operational principles • Q&A session • Clarification of doubts • Preparation for next lesson	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the role of the HEIU in maintaining hydraulic pressure.
(b) List three safety precautions to observe while working on the HEIU.

12. Home Assignment and Follow Up	Prepare a brief note on the differences between HEIU and other hydraulic units, referencing LP 02.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Demonstration of construction and explanation on operation of Synchro transmitters and receivers and Servo mechanisms.	Refer to the master syllabus reference list	LP 08

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min